

INTERACTIVE DEPTH COMPLETION: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

We study active perception for robotics. The motivating bet is: Complete depth by physically probing where perception uncertainty matters to action.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For active perception for robotics, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2025a; 2019; 2018; 2026; 2025b; 2024). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.550	0.450
observed_only_wam	hidden_mode_shift	0.185	0.815
augmentation_as_data_wam	nominal	0.568	0.432
augmentation_as_data_wam	hidden_mode_shift	0.253	0.748
scalar_uncertainty_wam	nominal	0.625	0.375
scalar_uncertainty_wam	hidden_mode_shift	0.323	0.677
proposed_branch_mechanism	nominal	0.738	0.262
proposed_branch_mechanism	hidden_mode_shift	0.588	0.412

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Interactive Depth Completion argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Model predictive control with active learning under model uncertainty: Why, when, and how, 2018. <https://doi.org/10.1002/aic.16180>.
- Learning guided convolutional network for depth completion, 2019. <https://doi.org/10.48550/arxiv.1908.01238>.
- Optimal and robust control methodologies for 4dof active suspension systems: A comparative study with uncertainty considerations, 2024. <https://doi.org/10.1002/oca.3192>.
- rt-riseg: Real-time model-free robot interactive segmentation for active instance-level object understanding, 2025a. <http://arxiv.org/abs/2507.10776v1>.
- Toward real-world bev perception: Depth uncertainty estimation via gaussian splatting, 2025b. <http://arxiv.org/abs/2504.01957v2>.
- Sapave: Towards active perception and manipulation in vision-language-action models for robotics, 2026. <http://arxiv.org/abs/2603.12193v1>.